MIL-R-3206B

8 July 1969

SUPERSEDING
MIL-R-3206A
17 June 1960

MILITARY SPECIFICATION

REELING MACHINE, CABLE, HAND, RL-39( )

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of light-weight, hand-reeling machine used for laying and recovering infantry field wire WD-1( )/TT; designated Reeling Machine RL-39( ). (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	Box, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-T-45	Tape, Gummed, Paper, Reinforced.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 3895

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MILITARY

MIL-P-116	Preservation, Methods of.
MIL-W-6858	Welding; Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys.
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sampling Procedure and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-454	Standard General Requirements for Electronic Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-333554	Reeling Machine RL-39( ).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Construction. Reeling Machine RL-39( ) shall be constructed in accordance with Drawing and Data List SC-DL-333554 and meet the requirements specified herein.

3.2 Parts, materials, and processes; general.

3.2.1 Plastic material and parts. Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

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3.2.2 Lubrication. Each reeling machine shall be lubricated with a few drops of SAE No. 20-W oil in the oil cover of both bearings and in the oil hole of the crank knob.

3.2.3 Riveting. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.2.4 Welding. Wherever practicable, welded joints shall conform to MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot welds shall conform to MIL-W-6858.

3.2.4.1 Cleaning prior to welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.2.4.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.2.4.3 Cleaning after welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.2.5 Tropicalization.

3.2.5.1 Treatment of material. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and shall be treated as specified herein.

3.2.5.2 Treating materials. Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compound shall be such that there shall be no increase in flammability of the treated material.

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3.2.5.3 Toxicity. Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.2.5.4 Flexibility. Treatment shall not affect the flexibility of the treated material, to the extent that the reeling machine may fail to meet specified requirements when subjected to specified service conditions.

3.2.5.5 Statement of treatment. The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2(e).)

3.2.6 Cleaning.

3.2.6.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.2.6.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.2.7 Finish. The reeling machine shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.3.)

3.2.8 Marking. Marking shall conform to MIL-M-13231. (See 4.3.)

3.3 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.7.)

3.4 Service conditions. The reeling machine shall meet the following service conditions.

3.4.1 Temperature. The reeling machine shall show no evidence of deterioration after being subjected to the temperature cycling test specified in 4.5.1, and shall meet the requirements of 3.5.1.

3.4.2 Moisture resistance. The reeling machine shall show no evidence of deterioration after being subjected to the moisture resistance test specified in 4.5.2, and shall meet the requirements of 3.5.1.

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3.4.3 Drop. The reeling machine shall show no physical damage except minor surface abrasion after being subjected to the test specified in 4.5.3, and shall meet the requirements of 3.5.1.

3.5 Operational requirements.

3.5.1 Operation. Reeling Machine RL-39( ) shall be capable of operation with Spool DR-8( ) complete with 1/4 mile of Wire WD-1( )/TT. (See 6.3.) The reeling machine with loaded DR-8( ) shall operate smoothly and without binding when cranked by hand to simulate recovery of wire. The bearings and shaft shall be examined visually to determine conformance. (See 4.6.1.)

3.5.2 Operational life (accelerated). After 240 hours of continuous cycling in accordance with 4.6.2, the reeling machine shall meet the requirements of 3.5.1 without adjustment or repair.

3.6 Workmanship. Reeling Machine RL-39( ) shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein: (See 4.8.)

Plastic materials and parts.
Lubrication.
Cleaning.
Riveting.
Welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3.)

(b) Quality conformance inspection.

(1) Quality conformance inspection of product before preparation for delivery. (See 4.4.)

(2) Quality conformance inspection of preparation for delivery. (See 4.9.)

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4.3 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Were required
Welding	3.2.4
Tropicalization	3.2.5
Finish	3.2.7
Marking	3.2.8

4.4 Quality conformance inspection of product before preparation for delivery. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection. Each unit (100 percent) of each lot of reeling machines shall be inspected for conformance to all inspection and test requirements of Table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using general inspection levels and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I. Group A inspection.

Examination or test	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical	3.6	4.8	1.0%	4.0%
Operation	3.5.1	4.6.1	6.5%	*

*Operational defects are considered major.

4.4.2 Group B inspection. This inspection including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be S-4. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and passed the group A inspection.

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Table II. Group B inspection.

Examination or test	Requirement paragraph	Inspection paragraph
Interchangeability	3.3	4.7

4.4.3 Group C inspection. This inspection shall consist of the tests specified in Table III and shall be performed on sample units that have been subjected to and passed group A and B inspections. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced. The moisture resistance and temperature test shall be performed one time only, at the beginning of production.

4.4.3.1 Sampling for inspection of equipment. Two units shall be selected from each 3,000 units or from every 2 months of production, whichever occurs first.

Table III. Group C inspection.

Examination or test	Requirement paragraph	Inspection paragraph
Temperature	3.4.1	4.5.1
Moisture resistance	3.4.2	4.5.2
Drop	3.4.3	4.5.3
Operational life (accelerated)	3.5.2	4.6.2

4.4.3.2 Noncompliance. The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

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4.4.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Service conditions tests.

4.5.1 Temperature. The reeling machine shall be subjected to the temperature cycle shown on MIL-STD-169. At the conclusion of the test, the reeling machine shall meet the requirements of 3.4.1.

4.5.2 Moisture resistance. The reeling machine shall be subjected to fifteen continuous 48-hour cycles per MIL-STD-170. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to the standard. After cycling has been completed, the reeling machine shall meet the requirements of 3.4.2.

4.5.3 Drop. The reeling machine shall be dropped from a height of 30 inches on each corner for a total of 8 drops. Drops shall be made from a quick-release hook, or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2 inch fir backed by concrete or a rigid steel frame. At the conclusion of the test, the reeling machine shall meet the requirements of 3.4.3.

4.6 Operational tests.

4.6.1 Operational. The reeling machine, loaded with Spool DR-8( ), shall be cranked by hand until the full capacity of the spool is loaded with Wire WD-1( )/TT. (See 6.3.) The operation of the reeling machine shall be observed during rotation, and shall conform to 3.5.1.

4.6.2 Operational life (accelerated). The reeling machine, containing Spool DR-8( ) fully loaded with Wire WD-1( )/TT (see 6.3), shall be cranked 50 to 60 revolutions per minute continuously for 240 hours, consisting of ten 24-hour cycles. The cranking may be stopped for a period not to exceed 15 minutes per 24-hour cycle for lubrication. At the conclusion of the test period, the reeling machine shall be disassembled and the steel bearings and the interior surface of their housing shall be inspected for conformance with 3.5.2.

4.7 Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.3. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Cross sectional dimensions of shaft.

(b) Dimensions of bearings and bearing housing.

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(c) Dimensions of spindle and knob.

(d) Dimensions of holes in the knob.

(e) Location of holes in bearing housing and plate.

4.8 Visual and mechanical. The reeling machine shall be examined for the defects listed in Table IV. (See 3.6.)

Table IV. Classification of visual and mechanical defects.

Classification	Defect
Major	Burrs, shaft end.
	Crack on shaft at bend.
	Missing cotter pin holes.
	Bent shaft.
	Knob fails to rotate freely.
	Knob shaft rivet improperly set.
	Missing oil hole in handle.
	Oversize of shaft in cross-section.
	Split roller.
	Broken or loose angle braces.
	Missing or broken grease fittings.
	Undersize bearing holes.
	Frozen bearings.
	Rust.
Minor	Clogged oil hole.
	Illegible marking on handle.
	Burrs on handle.

4.9 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.10 Rough handling (preparation for delivery). When the rough handling test, in accordance with MIL-P-116, is required by the contract (see 6.2(f)), the following inspection shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Operational	4.6.1
Visual and mechanical	4.8

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging of Reeling Machine RL-39( ) shall be level A or C as specified. (See 6.2(c).)

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5.1.1 Level A.

5.1.1.1 Cleaning. Each reeling machine shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each reeling machine shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Each reeling machine shall be packaged in accordance with item 1 and 2 of Figure 1 and Table V.

5.1.2 Level C. Each reeling machine shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing of Reeling Machine RL-39( ) shall be level A, B or C as specified. (See 6.2(c).)

5.2.1 Level A. Twenty reeling machines, packaged as specified in 5.1.1.4, shall be packed in accordance with items 3, 4, 5 and 6 of Figure 1 and Table V. Closure of item 4 shall be in accordance with the appendix of the box specification.

5.2.2 Level B. Twenty reeling machines, packaged as specified in 5.1.1.4, shall be packed as specified in 5.2.1.

5.2.3 Level C. Reeling machines packaged as specified in 5.1.2, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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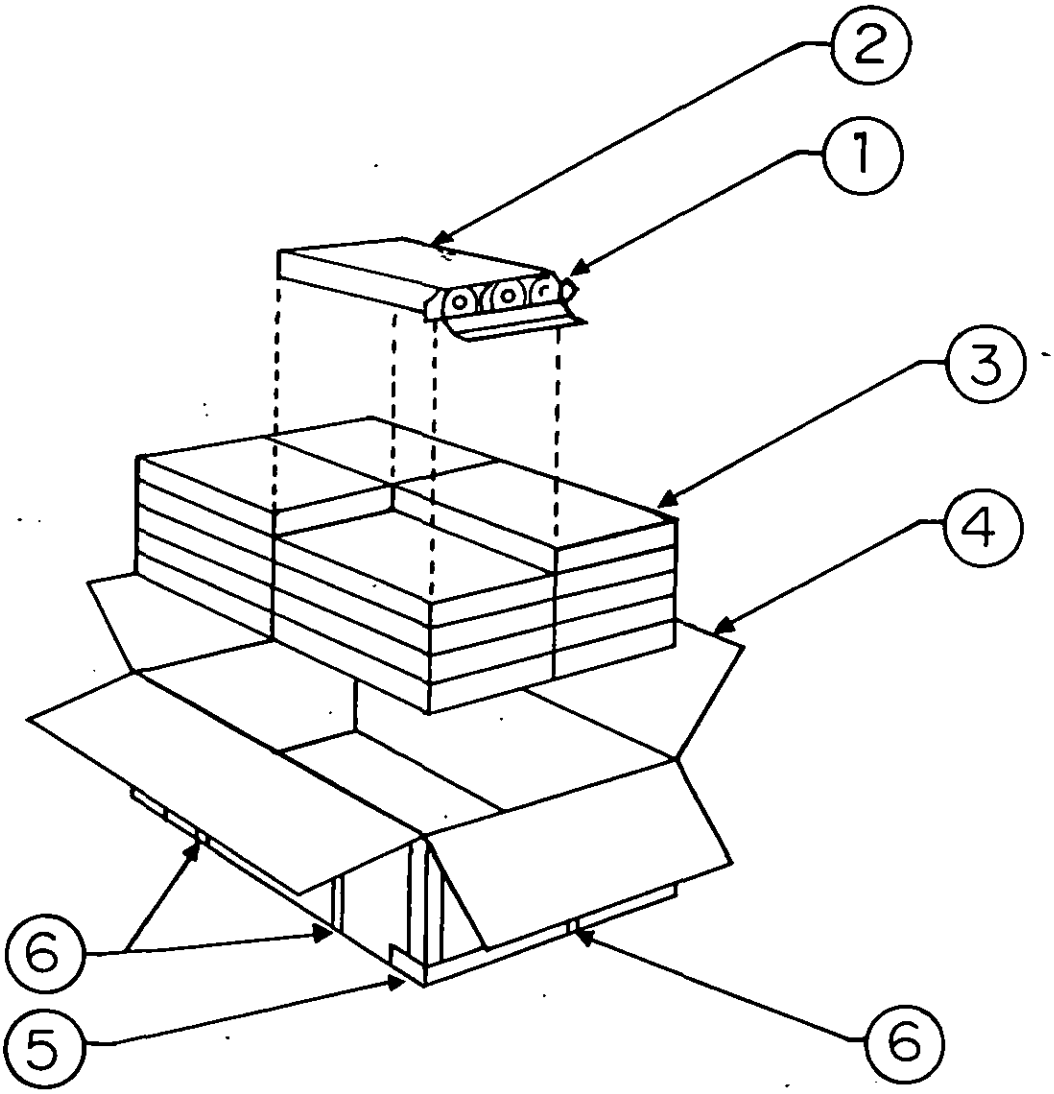


Figure 1. Packaging and packing of Reeling Machine RL-39( ).

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Table V. Bill of material.

Ref No.	Item Nomenclature	No. Reqd.	Method MIL-P-116	Size - Inches L	W	D	Spec No.	Requirement T	C	V	G	F	S
1	Reeling Machine RL-39( )	1	III										
	Crank, Reel	1	III										
	Cushion, Paper-board	1		14	9		PPP-P-291	I					I
	Tape, Paper Gummed	2		6	3		PPP-T-45	III			B		
	Reeling Unit	1	III										
	Cushion, Paper-board	1		30	6		PPP-P-291	I					I
	Tape, Paper Gummed	4		6	3		PPP-T-45	III			B		
	Strap ST-34	1	III										
	Cushion, Paper-board	1		10	10		PPP-P-291	I					I
	Tape, Paper Gummed	1		6	3		PPP-T-45	III			B		
	Strap ST-35	1	III										
	Cushion, Paper-board	1		10	10		PPP-P-291	I					I
	Tape, Paper Gummed	1		6	3		PPP-T-45	III			B		
2	Reeling Machine (Item 1 pkg)	1	III										
	Box, Folding	1		13-1/8	8	2-1/2	PPP-B-566	A	a	2			II
3	Reeling Machine (Item 2 Pkg)	20											
4	Box, Fiber	1		26-1/4	16	12-1/2	PPP-B-636	CF			V3c		
5	Tape, Pressure-Sensitive	2		33	3		PPP-T-76						
		4		22-1/2	3								
		1		19	3								
6	Tape, Reinforcing	1		80	1/2		PPP-T-97	III					
		2		60	1/2								

Lev. A - Packed weight 20 Units 76 lbs. T-Type V-Variety F-Flute
Lev. A - Packed volume 20 Units 3.3 cu. ft. C-Class G-Grade S-Style

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6. NOTES

6.1 Intended use. Reeling Machine RL-39( ) is a portable hand reeling device intended for laying and recovering infantry field wire when used with Spool DR-8( ).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required.

(c) Levels of preservation, packaging and packing. (See section 5.)

(d) The appropriate shipping container when other than a wood, cleated-plywood box is to be used as a shipping container.

(e) Submission of the statement of treatment referenced in 3.2.5.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

(f) Rough handling test, if required.

(g) Marking and shipping of samples.

(h) Place of final inspection.

6.3 Government-loaned material. Invitation for bids, contracts or other purchase documents should specify that the Government will loan 2 DR-8( ) Spools, fully loaded with field infantry wire WD-1( )/TT, for performing the operational tests. (See 3.5.1, 4.6.1 and 4.6.2.)

6.4 Dimensional data. Dimensions of packaging materials in the Bill of Material, Table V, are based on the dimensions of the reeling machine cited thereon. When the dimensions of the reeling machine vary, the dimensions of the packaging materials shall be adjusted accordingly.

6.5 Deviations. Requests for deviations shall be supported by test data proving the ability of the reeling machine, packaged and packed in accordance

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with the proposed design change, to withstand the rough handling test requirements of MIL-P-116. Approval will take into consideration the effects on standardization of techniques, availability of packaging and packing materials and economy.

Custodians:
Army - EL
Air Force - 84

Review Activities:
Navy - SH
DSA - CS

User Activities:
Army - ME
Navy - MC

Preparing Activity:
Army - EL

Project Nr. 3895-0194

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-3206A Reeling Machine, Cable, Hand, ML-39( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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